

September 2023 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains SEVEN questions. Answer ALL questions.

At the top of your answer scripts, please write the following:

- Your full name;
- Your class group; and
- Your class teacher's name.

Time Allowed: *2 hours*

You are supplied with: *Standard Normal and Student's t tables*

Calculators: *Calculators are NOT allowed in this examination*

1. Total: 15 points (18 minutes)

Let $\{A_1, A_2, \dots\}$ denote a set of events defined on sample space Ω . Let $\Pr(A_i)$ denote the probability of event A_i , and let A_i^c denote the complement of A_i . Let $\emptyset$ denote an empty set. Use the notation given above to answer the following questions.

- (a) State the axioms of probability. **(3pts)**
- (b) Prove that for any two events A_1 and A_2 ,

$$\Pr(A_1 \cup A_2) = \Pr(A_1) + \Pr(A_2) - \Pr(A_1 \cap A_2).$$

(6pts)

- (c) Now suppose we toss a fair coin twice. Let us define three events as follows:

A_1 : the first toss is heads,

A_2 : the second toss is heads, and

A_3 : the first and second tosses both show the same side of the coin.

Show that events A_1, A_2 and A_3 are pairwise independent.

(4pts)

- (d) Are events A_1, A_2 and A_3 (as defined in part (c)) fully independent?

(2pts)

2. Total: 15 points (18 minutes)

- (a) Consider the density function for random variable X given by

$$f_X(x) = \begin{cases} \alpha k^\alpha / x^{\alpha+1}, & \text{for } x \geq k, \\ 0, & \text{otherwise,} \end{cases}$$

where $\alpha > 1$ is an unknown parameter and $k > 0$ (the smallest possible value of X) is a known constant.

Derive the cumulative distribution function, $F_X(x)$.

(7pts)

- (b) Consider a different density function for random variable X given by

$$f_X(x) = \frac{1}{2} \exp(-|x|), \text{ for } -\infty < x < \infty.$$

Show that the moment-generating function of X is given by

$$M_X(t) = \frac{1}{(1 - t^2)},$$

and state the values of t for which $M_X(t)$ is well-defined.

(8pts)

3. Total: 10 points (12 minutes)

Consider the bivariate density function

$$f(x, y) = \begin{cases} \frac{3}{2}, & x^2 \leq y < 1, 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the conditional density function of Y given $X = x$. **(5pts)**
 - (b) Find $\Pr(X \geq 1/2, Y \geq 1/2)$. (You may leave your final answer in terms of simplified fractions.) **(5pts)**
-

4. Total: 15 points (18 minutes)

- (a) Consider an estimator $\hat{\mu}$ for unknown parameter μ . Define formally the statistical properties of bias, variance and mean-square error of $\hat{\mu}$. Derive an exact mathematical relationship between these three properties. **(5pts)**
- (b) Now suppose $\hat{\mu}_1$ and $\hat{\mu}_2$ are unbiased estimators of μ such that $\text{Var}(\hat{\mu}_1) = \sigma_1^2$ and $\text{Var}(\hat{\mu}_2) = \sigma_2^2$. A third estimator, $\hat{\mu}_3$, is proposed. It is defined as

$$\hat{\mu}_3 = k\hat{\mu}_1 + (1 - k)\hat{\mu}_2$$

for some constant k such that $0 < k < 1$.

- i. Evaluate the bias, variance and mean-square error of $\hat{\mu}_3$. **(3pts)**
 - ii. Suppose that $\hat{\mu}_1$ and $\hat{\mu}_2$ are independent. For what choice of k is $\hat{\mu}_3$ optimal in a minimum mean-square error sense? (You do **not** need to verify the second-order condition.) **(4pts)**
 - iii. Provide intuition for the expression you obtained in part ii. above. **(3pts)**
-

5. Total: 10 points (12 minutes)

Suppose we have a set of mutually independent random variables $Y_i \sim \text{Normal}(i\mu, i^3\sigma^2)$ for $i = 1, \dots, N$, where μ is unknown but σ^2 is known. Derive the maximum likelihood estimator for μ . (You do **not** need to verify the second-order condition.) **(10pts)**

Hint: You may find it helpful to recall that if $X \sim \text{Normal}(\mu, \sigma^2)$, then

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right) \text{ for any } x \in \mathbb{R}.$$

6. Total: 20 points (24 minutes)

Let X denote the lifespan (in years) of a given mouse. Assume that X is distributed $\text{Normal}(\mu, \sigma^2)$. Suppose σ^2 is unknown but it is known that $\mu = 3.4$ years. A scientist, Sara, claims that the mean lifespan of African pygmy mice is greater than 3.4 years. She plans to test her claim using a sample of 9 African pygmy mice.

- (a) Define H_0 and H_1 . Suggest a test statistic and state its distribution under H_0 . Are any additional assumptions needed? **(5pts)**
 - (b) Sara obtains sample average, $\bar{x} = 3.7$, and sample variance, $s^2 = 1/9$.
 - i. Given a 5% level of significance, discuss whether Sara can reject the null or not. Interpret the finding in your own words. **(5pts)**
 - ii. What is the p -value of the test? Give your best approximation. **(5pts)**
 - iii. If the true mean lifespan of African pygmy mice is 3.8 years, what is the power of the test? Give your best approximation. **(5pts)**
-

7. Total: 15 points (18 minutes)

- (a) Suppose you wish to choose a probability distribution as an approximating model for a real-world variable. Briefly describe what characteristics of the real-world variable you would take into account when making your choice. **(3pts)**
- (b) Let X and Y represent two continuous random variables, and let x denote an arbitrary realised value of X .
 - i. Explain, in your own words, the difference between $E[Y|X]$ and $E[Y|X = x]$. Do these terms represent random variables or fixed parameters? **(4pts)**
 - ii. Suppose that $E(Y|X = x) = \beta_0 + \beta_1 x$, where the β 's represent fixed parameters. State the law of iterated expectations (LIE). Using the LIE, or otherwise, prove that

$$\text{Corr}(X, Y) = \beta_1 \sqrt{\frac{\text{Var}(X)}{\text{Var}(Y)}}$$

and comment on this finding. **(8pts)**

TABLE FD.1 Observations on Income and Education, 20 Observations

I = Observation,
 Y = Income.

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z < z]$

[illegible]

TABLE G.2 Percentiles of the Student's t Distribution. Table Entry Is x Such that $\text{Prob}[t_n \leq x] = P$

n	.750	.900	.950	.975	.990	.995
1	1.000	3.078	6.314	12.706	31.821	63.657
2	.816	1.886	2.920	4.303	6.965	9.925
3	.765	1.638	2.353	3.182	4.541	5.841
4	.741	1.533	2.132	2.776	3.747	4.604
5	.727	1.476	2.015	2.571	3.365	4.032
6	.718	1.440	1.943	2.447	3.143	3.707
7	.711	1.415	1.895	2.365	2.998	3.499
8	.706	1.397	1.860	2.306	2.896	3.355
9	.703	1.383	1.833	2.262	2.821	3.250
10	.700	1.372	1.812	2.228	2.764	3.169
11	.697	1.363	1.796	2.201	2.718	3.106
12	.695	1.356	1.782	2.179	2.681	3.055
13	.694	1.350	1.771	2.160	2.650	3.012
14	.692	1.345	1.761	2.145	2.624	2.977
15	.691	1.341	1.753	2.131	2.602	2.947
16	.690	1.337	1.746	2.120	2.583	2.921
17	.689	1.333	1.740	2.110	2.567	2.898
18	.688	1.330	1.734	2.101	2.552	2.878
19	.688	1.328	1.729	2.093	2.539	2.861
20	.687	1.325	1.725	2.086	2.528	2.845
21	.686	1.323	1.721	2.080	2.518	2.831
22	.686	1.321	1.717	2.074	2.508	2.819
23	.685	1.319	1.714	2.069	2.500	2.807
24	.685	1.318	1.711	2.064	2.492	2.797
25	.684	1.316	1.708	2.060	2.485	2.787
26	.684	1.315	1.706	2.056	2.479	2.779
27	.684	1.314	1.703	2.052	2.473	2.771
28	.683	1.313	1.701	2.048	2.467	2.763
29	.683	1.311	1.699	2.045	2.462	2.756
30	.683	1.310	1.697	2.042	2.457	2.750
35	.682	1.306	1.690	2.030	2.438	2.724
40	.681	1.303	1.684	2.021	2.423	2.704
45	.680	1.301	1.679	2.014	2.412	2.690
50	.679	1.299	1.676	2.009	2.403	2.678
60	.679	1.296	1.671	2.000	2.390	2.660
70	.678	1.294	1.667	1.994	2.381	2.648
80	.678	1.292	1.664	1.990	2.374	2.639
90	.677	1.291	1.662	1.987	2.368	2.632
100	.677	1.290	1.660	1.984	2.364	2.626
∞	.674	1.282	1.645	1.960	2.326	2.576